

ICT200
Introduction to Database Management

Final Report

Retail Shop Database System
Pasar Mini Mohd. Salleh

SEMESTER OCTOBER 2025 – FEBRUARY 2026

GROUP: CS1103C

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DATE: 23/1/2025

[A] Final Report

1	2	3	4	5
Poor	Below Average	Average	Good	Excellent
Unacceptable. Performance was <u>below</u> expectations. Needs significant improvement.	The student attempted but did <u>not</u> achieve competency in all areas. Needs improvement.	The student performed the competence at an <u>acceptable</u> level. Assistance or guidance from the preceptor was occasionally necessary.	. The student performed the competency very well, was <u>above average</u> in effectiveness, and/or consistency.	The student has excelled in performing competencies, works independently. The student has completed the work and <u>exceeded</u> activity requirements.

Criteria	Score					Weight	Total Score	Total / Section
	1	2	3	4	5	w	w x score	
Report Format	Report poorly formatted; inconsistent layout	Some formatting inconsistencies; lacks clarity	Adequately formatted. Minor layout issues	Well formatted with clear headings and spacing	Professional, consistent, and polished formatting	0.6		3
Company background	Missing or irrelevant company information	Superficial or confusing description	Basic background info is present and understandable	Clear and concise with relevant company details	Comprehensive, insightful background with context	1		5
Daily Operation Description	Operations are poorly described or missing	Overview of operations unclear or incomplete	Basic description of operations	Clear description with sufficient detail	Thorough, detailed and well-structured explanation. Supported with related diagrams	1		5
Problem Findings	The problem statement is missing or vague. No specific data management problems were described.	The problem statement lists fewer than 3 problems or is only superficially related to data management.	Clearly describes 3 data management problems but lacks depth or relevance in explanation.	Well-articulated problem identifies 3 relevant data management problems with reasonable detail	Comprehensive problem statements with at least 3 clearly defined, relevant, and well-explained data management problems demonstrate a clear understanding of impact	1		5
Project Objectives	Objectives are missing or unclear	Objectives vague or incomplete	Objectives stated but lacking detail	Clear and well-defined objectives	Precise, comprehensive, and aligned with	1		5

(A1) Rubric of Final Report

Criteria	Score					Weight	Total Score	Total / Section
	1	2	3	4	5	w	w x score	
Report Format	Report poorly formatted; inconsistent layout	Some formatting inconsistencies; lacks clarity	Adequately formatted. Minor layout issues	Well formatted with clear headings and spacing	Professional, consistent, and polished formatting	0.6		3
Company background	Missing or irrelevant company information	Superficial or confusing description	Basic background info is present and understandable	Clear and concise with relevant company details	Comprehensive, insightful background with context	1		5
Daily Operation Description	Operations are poorly described or missing	Overview of operations unclear or incomplete	Basic description of operations	Clear description with sufficient detail	Thorough, detailed and well-structured explanation. Supported with related diagrams	1		5
Problem Findings	The problem statement is missing or vague. No specific data management problems were described.	The problem statement lists fewer than 3 problems or is only superficially related to data management.	Clearly describes 3 data management problems but lacks depth or relevance in explanation.	Well-articulated problem identifies 3 relevant data management problems with reasonable detail	Comprehensive problem statements with at least 3 clearly defined, relevant, and well-explained data management problems demonstrate a clear understanding of impact	1		5
Project Objectives	Objectives are missing or unclear	Objectives vague or incomplete	Objectives stated but lacking detail	Clear and well-defined objectives	Precise, comprehensive, and aligned with	1		5

(A2) Rubric of Database Design

Final ERD – using Crow's Foot notation	ERD is incomplete. Attributes are missing, keys are not identified, and relationships are incorrect or missing. Crow's Foot notation is not used or used incorrectly	ERD shows some entities, but many attributes or key attributes are missing or incorrectly marked. Relationships are poorly defined, and Crow's Foot notation is inconsistently applied	ERD has most entities with attributes and keys identified correctly. Relationships are mostly Correct with Crow's Foot notation used properly, but some minor mistakes or omissions	ERD shows complete entities with attributes and keys clearly identified. Relationships are properly defined with the correct use of Crow's Foot notation, indicating cardinality and modality accurately. Minor details might be missing	ERD is fully complete with all attributes and keys correctly included. All relationships between entities are thoroughly defined and clearly shown with accurate Crow's Foot notation, including cardinality, modality, and integrity constraints. Diagrams are neat and easy to understand	1	5
Normalized ERD	ERD shows no signs of normalization. Tables have clear redundant fields or anomalies	Partial normalization but some redundant or repeated data remains	ERD shows basic normalization (1NF and 2NF) but may miss some refinements	ERD is mostly normalized with minimal redundancy. Most anomalies addressed	ERD fully normalized (up to 3NF as required), eliminating redundancy and ensuring data integrity	1	5
Data Dictionary	The data dictionary is missing or incomplete with many entries omitted or incorrect	Data dictionary has partial entries; some attribute definitions or data types are unclear or incorrect	The data dictionary includes all entities and attributes, but definitions or data types may be inconsistent	Data dictionary clearly defines entities, attributes and data types with few errors	Complete and well-organized data dictionary clearly defining all entities, attributes, data types, constraints and additional notes	1	5

(A3) Rubric of Database Implementation: DDL & DML

Criteria	Score					Weight	Total Score	Total / Section
	1 (POOR)	2 (BELOW AVERAGE)	3 (AVERAGE)	4 (GOOD)	5 (EXCELLENT)	w	w x score	
Execution of DDL Statements	DDL statements are mostly incorrect or missing	Some correct statements but with major errors	Basic correct DDL commands with minor mistakes	Accurate statements with few minor issues.	Fully correct, well-structured DDL with all required elements.	1		5
Execution of DML for Single Table Data Retrieval	DML queries are incorrect or missing	Simple queries with many syntax or logic errors	Queries retrieve data with minor mistakes or inefficiencies	Correct queries that retrieve relevant data properly	Efficient and accurate queries demonstrating good SQL use	1		5
Execution of DML for Multi-table Retrieval Using Different Relationship Clauses	No multi-table queries or completely incorrect	Queries attempt joins but contain errors	Basic join queries with some minor issues	Correct join queries covering multiple relationship types	Complex, accurate joins demonstrating a thorough understanding	1		5
Execution of DML Using Comparison Operators for Data Filtering	No use or incorrect use of comparison operators.	Basic filtering but syntax or logic errors are present.	Functional use of comparison operators with minor errors.	Correct and efficient use of operators for filtering.	Sophisticated filtering demonstrating mastery of operators.	1		5
Execution of DML Using Comprehensive Column Functions	Functions are missing or used incorrectly	Basic use with errors.	Correct use of some scalar or aggregate functions	Efficient use of column functions covering multiple cases	Advanced and correct use of functions optimizing queries	2		10
Execution of DML Using Comprehensive Subqueries	No subqueries or incorrect usage.	Attempts at subqueries but with errors.	Functionally correct subqueries with minor flaws	Accurate and effective use of subqueries.	Complex and well-optimized subqueries demonstrate deep understanding.	1		5
Relevance of Record in Each Table for Data Manipulation	Records are mostly irrelevant or unrelated to the table's purpose; data manipulation queries result in incorrect or meaningless outputs.	Many records are relevant but there are significant irrelevant or redundant entries affecting query outcomes.	Most records are relevant and support intended data operations; minor inconsistencies may exist.	Records are carefully selected and consistent, enabling accurate and meaningful data manipulation with negligible irrelevant data.	All records are highly relevant and precisely aligned with the table's role; queries produce consistently correct and valuable results.	0.6		3
TOTAL MARKS FOR PROJECT REPORT								76

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1.0 Members Profile

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2.0 Company Background

PASAR MINI MOHD. SALLEH operates not as a typical commercial retail shop, but as a critical service provider deeply embedded within the ecosystem of UiTM campus Jasin. It's strategic location likely within a residential college, student centre, or central academic block positions it as the go-to hub for the immediate needs of the campus population, particularly students living in residence halls.

Its business model mimics a modern convenience store (like a 7-Eleven), focusing on speed, extended operational hours, and a very specific product mix: ready-to-eat and ready-to-drink items, quick snacks, personal necessities, and essential academic supplies (e.g., specific stationery, printing cards). The store's success is intrinsically tied to the university's rigid academic calendar and class schedule.

VISION :

To be the most efficient, reliable, and essential retail convenience store for the entire university community, setting the standard for intra-campus retail service and operational excellence.

MISSION :

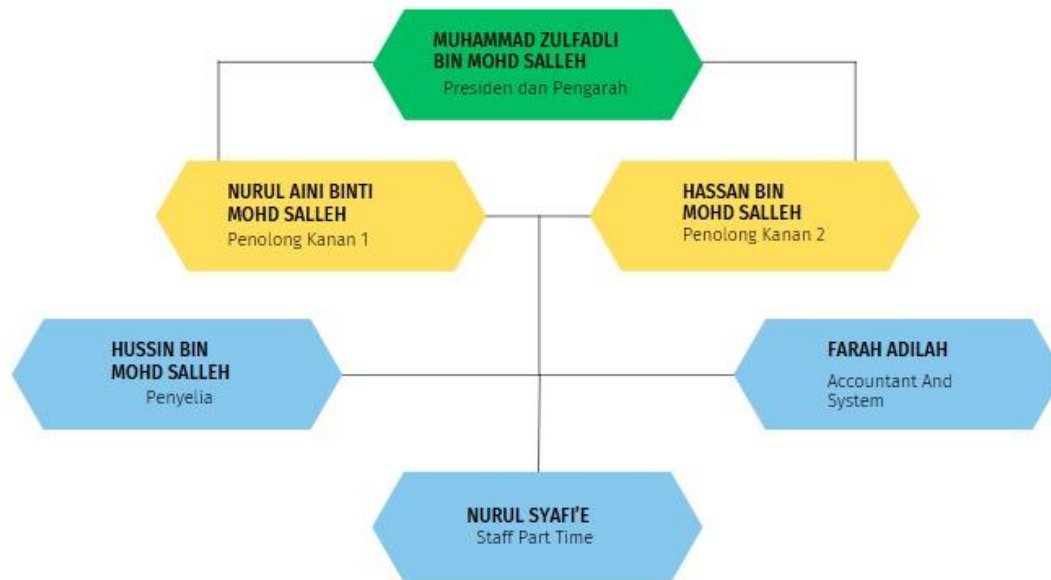
To provide swift, convenient, and high-quality access to daily necessities and essential goods, ensuring optimal stock availability while maintaining rapid transaction speeds that support the demanding schedules of students and faculty.

OBJECTIVE :

1. Maximize Convenience: To minimize the time students and staff spend on purchasing necessities.
2. Optimize Operations: To maintain a lean operational model that ensures product availability while minimizing inventory waste (spoilage and overstocking).
3. Enhance Service Speed: To ensure quick and error-free transactions, especially during peak rush hours.
4. Support Campus Life: To function as a reliable resource that contributes positively to the overall student life experience.

ORGANIZATIONAL CHART :

PASAR MINI MOHD SALLEH



3.0 Description of Daily Operation and Data Needs

1. Customer File

- **Purpose:** Maintain records of recurring customers, loyalty accounts, and transaction-linked customer details to support marketing, returns, and dispute resolution.
- **Key fields:** Customer ID, Customer Name, Customer Gender, Customer Phone Number, Customer Email
- **Process (step by step):**
 - Register new customers and assign a Customer ID.
 - Link purchases to Customer ID during transactions.
 - Update customer details when changes occur.
 - Use customer data for loyalty programs and marketing reports.
 - Maintain privacy and delete data according to company policy.

2. Product File

- **Purpose:** Central master file for every sellable unit; used for pricing, shelf stocking, expiry control, and reorder decisions.
- **Key fields:** Product ID, Product Name, Product Category, Product Brand, Product Cost Price, Product Sell Price
- **Process (step by step):**
 - Create new product entries with full details.
 - Update stock when new goods are received.
 - Deduct stock automatically after each sale.
 - Check for low stock to generate reorder alerts.
 - Conduct physical stock counts and record any adjustments.

3. Supplier File

- **Purpose:** Maintain vendor details for procurement, billing, contact, and supplier performance tracking.
- **Key fields:** Supplier ID, Supplier Name, Supplier Email, Supplier Unit Number, Supplier Street, Supplier Neighborhood, Supplier Town, Supplier State
- **Process (step by step):**
 - Register new suppliers and assign Supplier ID.
 - Link Purchase Orders to the correct supplier.
 - Verify received goods and match with supplier records.
 - Review supplier performance regularly.
 - Update supplier details when information changes.

4. Sales File

- **Purpose:** Record every point-of-sale event for revenue tracking, cashier accountability, inventory decrement, and customer receipts.
- **Key fields:** Sales ID, Sales Date Time, Sales POS, Sales Total Amount, Sales Discounts, Sales Final Total, Sales Status
- **Process (step by step):**
 - Record each sale at the POS terminal.
 - Link transaction to cashier and customer (if applicable).
 - Update inventory quantities automatically.
 - Generate receipts and process returns when needed.
 - Summarize sales data for daily and monthly reports.

5. Employee File

- **Purpose:** Maintain personnel records for scheduling, payroll, access control, and transaction accountability.
- **Key fields:** Employee ID, Employee Name, Employee Role, Employee Phone Number, Employee Status, Employee Start Date, Employee Work Hours, Employee Certifications, Employee Salary
- **Process (step by step):**
 - Add new employees with assigned ID and role.
 - Record attendance and work schedules.
 - Link employee IDs to sales or purchase activities.
 - Use data for payroll and performance review.
 - Update records when employees leave or change roles.

6. Expense File

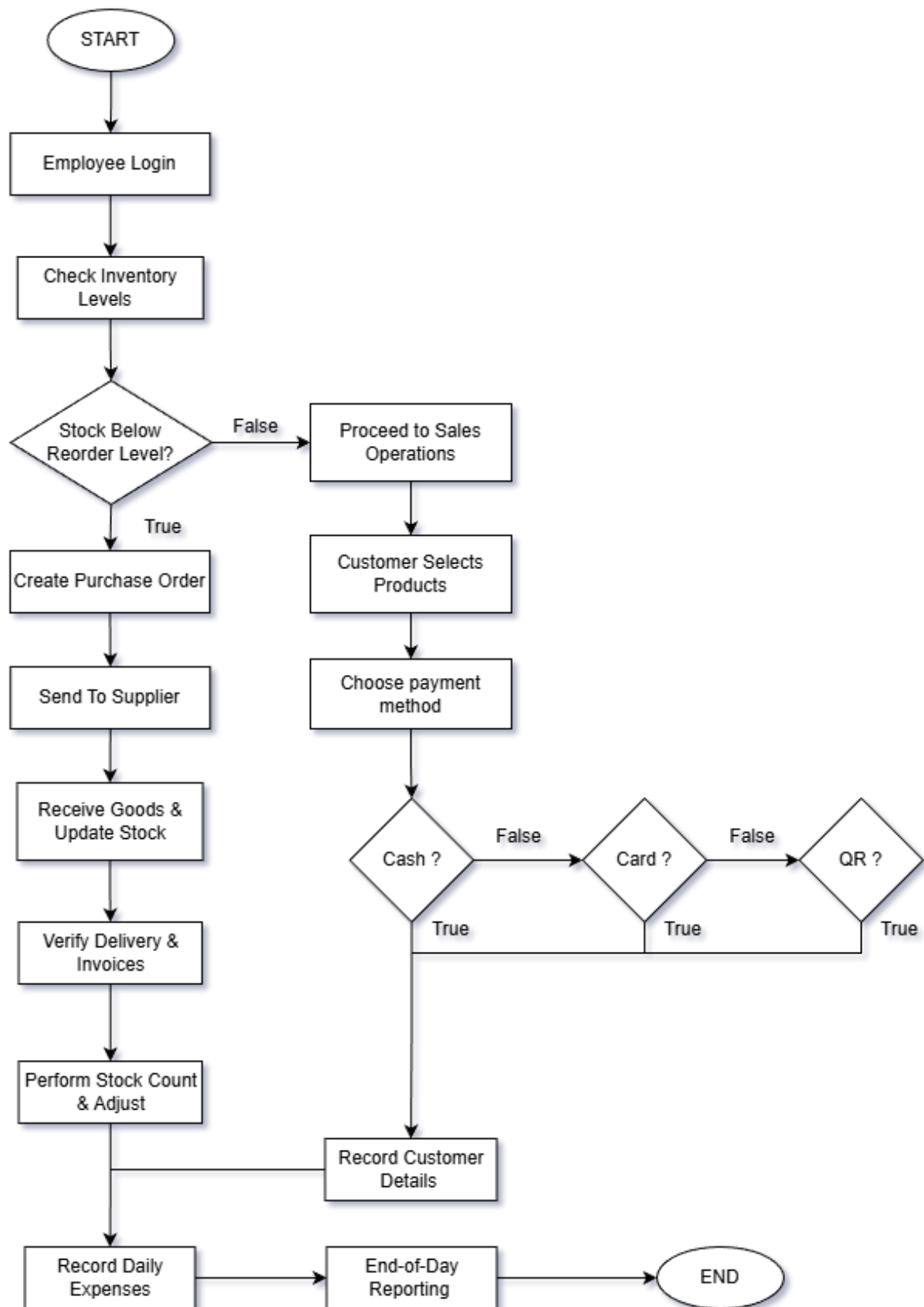
- **Purpose:** Capture non-inventory business expenditures (utilities, rent, petty cash, repairs) for financial control and profitability analysis.
- **Key fields:** Expense ID, Expense Date, Expense Amount, Expense Type, Expense Status, Expense Category, Expense Description
- **Process (step by step):**
 - Record expenses when bills or receipts are received.
 - Approve and verify expenses before payment.
 - Process payments and update financial records.
 - Include expenses in profit and loss reports.
 - Keep receipts for audit and verification.

7. Stock File

- **Purpose:** Record manual or system-triggered adjustments to inventory quantities due to shrinkage, damage, theft, count corrections, or other reconciliation activities.
- **Key fields:** Stock ID, Stock Quantity, Stock Unit Cost, Stock Total Cost, Stock Status
- **Process (step by step):**
 - Identify stock discrepancies or damages.
 - Record adjustment details and reason.
 - Approve adjustments before applying changes.
 - Update inventory with new quantity.
 - Include adjustment data in loss or audit reports.

8. Payment File

- **Purpose:** Record all payment transactions linked to sales, track payment methods, amounts, and statuses to ensure accurate financial reporting and customer transaction verification.
- **Key fields:** Payment ID, Payment Method, Payment Amount, Payment Date, Payment Status
- **Process (step by step):**
 - Record each payment made for a sale at the POS terminal.
 - Link the payment to the corresponding Transaction ID.
 - Verify the payment amount matches the total due.
 - Update payment status (e.g., Completed, Pending, Failed, Refunded).
 - Use payment data for daily cash reconciliation and accounting reports.
 - Keep records for audits and customer dispute resolution.



4.0 Problem Findings in Data Management

- Lengthy Development Time

Manual file systems require more time to create the needed database as they use only a piece of paper and pencil. The employees who are in charge at that moment must write down the records that have been received on paper each time. Right before the end of the day, they must put the paper in a file. Different files contain different records such as employee files, customer files and etc. Besides that, to differentiate it from the years, they have to use a different file. This not only slows down the development timeout, also slows down the searching time.

- Data Entry Error

As they have to record the data manually on paper, human errors are unavoidable. As a consequence, the data integrity could not be trusted. For example, the number of products that have been sold are more or less than the data recorded. This creates a confusion in the retail which would make the retail owner forget to restock the products. So, there is an existence of expired goods.

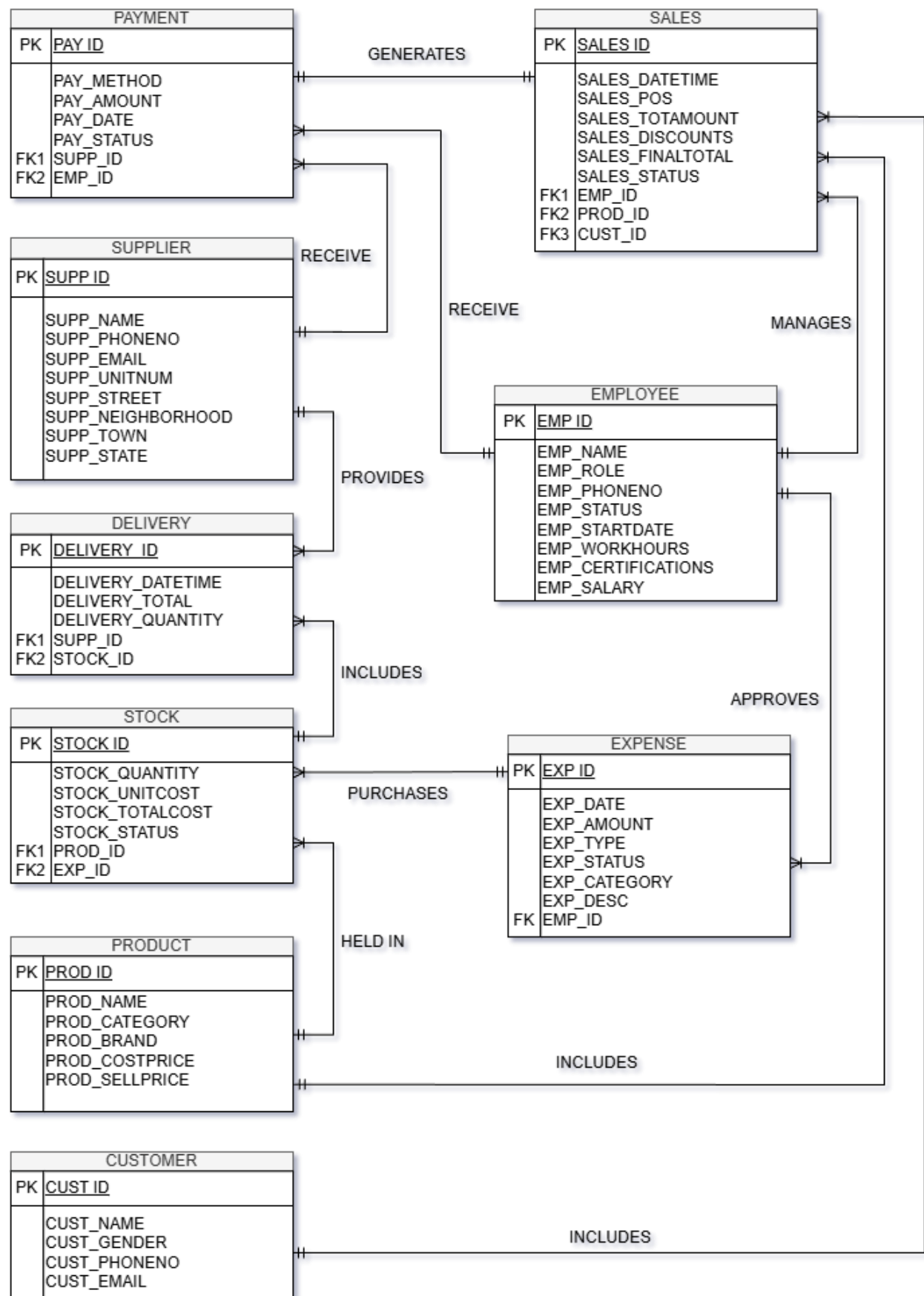
- Lack of security and limited data sharing

The manual file systems in the retail shop don't have any security mechanism to protect the records. As of this, the records are vulnerable and can be altered by anyone even strangers. The most worrying part, the employees could change the data in the record to match the amount of money they could steal and no one would know. This is because the manual file systems do not have constraints or an auto deduct product each time the products are scanned. As a result, data sharing between employees is limited, and data integrity is easily compromised.

5. 0 Objectives of the Database System

1. To clearly understand what the company needs in order to manage its data properly, especially to avoid problems like missing files, slow searching, and weak data security that affect daily work.
2. To design a database that keeps all information safe by using automatic backups and strong security features, while also making it easy for staff to find information quickly without digging through paperwork.
3. To build a system that supports the company's routine tasks smoothly, reduces the chance of losing important data, speeds up work processes, and helps the management make better decisions using accurate and up-to-date information.
4. To create a central place where all company data is stored and organized, making it easier for different departments to update, access, and share information efficiently.

6.0 Finalized Entity Relationship Diagram



7.0 Normalized ERD

EMPLOYEE (EMP_ID, EMP_NAME, EMP_ROLE, EMP_PHONENO, EMP_STATUS, EMP_STARTDATE, EMP_WORKHOURS, EMP_CERTIFICATIONS, EMP_SALARY)

CUSTOMER (CUST_ID, CUST_NAME, CUST_GENDER, CUST_PHONENO, CUST_EMAIL)

PRODUCT (PROD_ID, PROD_NAME, PROD_CATEGORY, PROD_BRAND, PROD_COSTPRICE, PROD_SELLPRICE)

SALES (SALES_ID, SALES_DATETIME, SALES_POS, SALES_TOTAMOUNT, SALES_DISCOUNTS, SALES_FINALTOTAL, SALES_STATUS, EMP_ID*, PROD_ID*, CUST_ID*)

PAYMENT (PAY_ID, PAY_METHOD, PAY_AMOUNT, PAY_DATE, PAY_STATUS, SUPP_ID*, EMP_ID*)

SUPPLIER (SUPP_ID, SUPP_NAME, SUPP_PHONENO, SUPP_EMAIL, SUPP_UNITNUM, SUPP_STREET, SUPP_NEIGHBORHOOD, SUPP_TOWN, SUPP_STATE)

DELIVERY (DELIVERY_ID, DELIVERY_DATETIME, DELIVERY_TOTAL, DELIVERY_QUANTITY, SUPP_ID*, STOCK_ID*)

STOCK (STOCK_ID, STOCK_QUANTITY, STOCK_UNITCOST, STOCK_TOTALCOST, STOCK_STATUS, PROD_ID*, EXP_ID*)

EXPENSE (EXP_ID, EXP_DATE, EXP_AMOUNT, EXP_TYPE, EXP_STATUS, EXP_CATEGORY, EXP_DESC, EMP_ID*)

8.0 Data Dictionary

Table Name	Attribute Name	Datatype	Nullable	PK/ FK	Reference Table
EMPLOYEE	<u>EMP_ID</u>	INT	No	PK	
	EMP_NAME	VARCHAR(100)	No		
	EMP_ROLE	VARCHAR(50)	No		
	EMP_PHONENO	VARCHAR(20)	Yes		
	EMP_STATUS	VARCHAR(20)	No		
	EMP_STARTDATE	DATE	No		
	EMP_WORKHOURS	DECIMAL(5,2)	Yes		
	EMP_CERTIFICATIONS	TEXT	Yes		
	EMP_SALARY	DECIMAL(10,2)	No		
CUSTOMER	<u>CUST_ID</u>	INT	No	PK	
	CUST_NAME	VARCHAR(100)	No		
	CUST_GENDER	CHAR(1)	Yes		
	CUST_PHONENO	VARCHAR(20)	No		
	CUST_EMAIL	VARCHAR(100)	No		
PRODUCT	<u>PROD_ID</u>	INT	No	PK	
	PROD_NAME	VARCHAR(100)	No		
	PROD_CATEGORY	VARCHAR(50)	Yes		
	PROD_BRAND	VARCHAR(50)	Yes		
	PROD_COSTPRICE	DECIMAL(10,2)	No		
	PROD_SELLPRICE	DECIMAL(10,2)	No		
DELIVERY	<u>DELIVERY_ID</u>	INT	No	PK	
	DELIVERY_DATETIME	DATETIME	No		
	DELIVERY_TOTAL	DECIMAL(10,2)	No		
	DELIVERY_QUANTITY	INT	No		
	SUPP_ID*	INT	No	FK	SUPPLIER
	STOCK_ID*	INT	No	FK	STOCK

SALES	<u>SALES_ID</u>	INT	No	PK	
	SALES_DATETIME	DATETIME	No		
	SALES_POS	VARCHAR(50)	Yes		
	SALES_TOTAMOUNT	DECIMAL(10,2)	No		
	SALES_DISCOUNTS	DECIMAL(10,2)	Yes		
	SALES_FINALTOTAL	DECIMAL(10,2)	No		
	SALES_STATUS	VARCHAR(20)	No		
	EMP_ID*	INT	No	FK	EMPLOYEE
	PROD_ID*	INT	No	FK	PRODUCT
	CUST_ID*	INT	No	FK	CUSTOMER
PAYMENT	<u>PAY_ID</u>	INT	No	PK	
	PAY_METHOD	VARCHAR(50)	No		
	PAY_AMOUNT	DECIMAL(10,2)	No		
	PAY_DATE	DATE	No		
	PAY_STATUS	VARCHAR(20)	No		
	SUPP_ID*	INT	Yes	FK	SUPPLIER
	EMP_ID*	INT	No	FK	EMPLOYEE
STOCK	<u>STOCK_ID</u>	INT	No	PK	
	STOCK_QUANTITY	INT	No		
	STOCK_UNITCOST	DECIMAL(10,2)	No		
	STOCK_TOTALCOST	DECIMAL(10,2)	No		
	STOCK_STATUS	VARCHAR(20)	No		
	PROD_ID*	INT	No	FK	PRODUCT
	EXP_ID*	INT	Yes	FK	EXPENSE

EXPENSE	<u>EXP_ID</u>	INT	No	PK	
	EXP_DATE	DATE	No		
	EXP_AMOUNT	DECIMAL(10,2)	No		
	EXP_TYPE	VARCHAR(50)	No		
	EXP_STATUS	VARCHAR(20)	No		
	EXP_CATEGORY	VARCHAR(50)	Yes		
	EXP_DESC	TEXT	Yes		
	EMP_ID*	INT	No	FK	EMPLOYEE
SUPPLIER	SUPP_ID	INT	No	PK	
	SUPP_NAME	VARCHAR(100)	No		
	SUPP_PHONENO	VARCHAR(20)	No		
	SUPP_EMAIL	VARCHAR(100)	Yes		
	SUPP_UNITNUM	VARCHAR(20)			
	SUPP_STREET	VARCHAR(100)			
	SUPP_NEIGHBORHOOD	VARCHAR(50)			
	SUPP_TOWN	VARCHAR(50)			
	SUPP_STATE	VARCHAR(50)			

9.0 Execution of Data Definition Language (DDL) Statements

9.1 Create and eliminate the database

Creating the Database

```
CREATE DATABASE retail_shop;
```

Eliminating the Database

```
DROP DATABASE retail_shop;
```

9.2 Create table commands for all tables

```
CREATE TABLE employee (  
    emp_id INT AUTO_INCREMENT PRIMARY KEY,  
    emp_name VARCHAR(100) NOT NULL,  
    emp_role VARCHAR(50),  
    emp_phoneno VARCHAR(20),  
    emp_status VARCHAR(20),  
    emp_startdate DATE,  
    emp_workhours DECIMAL(5,2),  
    emp_certifications TEXT,  
    emp_salary DECIMAL(10,2)  
);  
  
CREATE TABLE customer (  
    cust_id INT AUTO_INCREMENT PRIMARY KEY,  
    cust_name VARCHAR(100) NOT NULL,  
    cust_gender CHAR(1),  
    cust_phoneno VARCHAR(20),  
    cust_email VARCHAR(100)  
);  
  
CREATE TABLE product (  
    prod_id INT AUTO_INCREMENT PRIMARY KEY,  
    prod_name VARCHAR(100) NOT NULL,  
    prod_category VARCHAR(50),  
    prod_brand VARCHAR(50),  
    prod_costprice DECIMAL(10,2),  
    prod_sellprice DECIMAL(10,2)  
);
```

```

CREATE TABLE supplier (
    supp_id INT AUTO_INCREMENT PRIMARY KEY,
    supp_name VARCHAR(100) NOT NULL,
    supp_phoneno VARCHAR(20),
    supp_email VARCHAR(100),
    supp_unitnum VARCHAR(20),
    supp_street VARCHAR(100),
    supp_neighborhood VARCHAR(50),
    supp_town VARCHAR(50),
    supp_state VARCHAR(50)
);

CREATE TABLE expense (
    exp_id INT AUTO_INCREMENT PRIMARY KEY,
    exp_date DATE,
    exp_amount DECIMAL(10,2),
    exp_type VARCHAR(50),
    exp_status VARCHAR(20),
    exp_category VARCHAR(50),
    exp_desc TEXT,
    emp_id INT,
    FOREIGN KEY (emp_id) REFERENCES employee(emp_id)
);

CREATE TABLE stock (
    stock_id INT AUTO_INCREMENT PRIMARY KEY,
    stock_quantity INT,
    stock_unitcost DECIMAL(10,2),
    stock_totalcost DECIMAL(10,2),
    stock_status VARCHAR(20),
    prod_id INT,
    exp_id INT,
    FOREIGN KEY (prod_id) REFERENCES product(prod_id),
    FOREIGN KEY (exp_id) REFERENCES expense(exp_id)
);

```

```

CREATE TABLE payment (
    pay_id INT AUTO_INCREMENT PRIMARY KEY,
    pay_method VARCHAR(50),
    pay_amount DECIMAL(10,2),
    pay_date DATE,
    pay_status VARCHAR(20),
    supp_id INT,
    emp_id INT,
    FOREIGN KEY (supp_id) REFERENCES supplier(supp_id),
    FOREIGN KEY (emp_id) REFERENCES employee(emp_id)
);

```

```

CREATE TABLE sales (
    sales_id INT AUTO_INCREMENT PRIMARY KEY,
    sales_datetime DATETIME,
    sales_pos VARCHAR(50),
    sales_totamount DECIMAL(10,2),
    sales_discounts DECIMAL(10,2),
    sales_finalltotal DECIMAL(10,2),
    sales_status VARCHAR(20),
    emp_id INT,
    prod_id INT,
    cust_id INT,
    FOREIGN KEY (emp_id) REFERENCES employee(emp_id),
    FOREIGN KEY (prod_id) REFERENCES product(prod_id),
    FOREIGN KEY (cust_id) REFERENCES customer(cust_id)
);

```

```

CREATE TABLE delivery (
    delivery_id INT AUTO_INCREMENT PRIMARY KEY,
    delivery_datetime DATETIME,
    delivery_total DECIMAL(10,2),
    delivery_quantity INT,
    supp_id INT,
    stock_id INT,
    FOREIGN KEY (supp_id) REFERENCES supplier(supp_id),
    FOREIGN KEY (stock_id) REFERENCES stock(stock_id)
);

```


9.3 Create two tables that have a relationship between them

```
CREATE TABLE employee (  
    emp_id INT AUTO_INCREMENT PRIMARY KEY,  
    emp_name VARCHAR(100) NOT NULL,  
    emp_role VARCHAR(50),  
    emp_phoneno VARCHAR(20),  
    emp_status VARCHAR(20),  
    emp_startdate DATE,  
    emp_workhours DECIMAL(5,2),  
    emp_certifications TEXT,  
    emp_salary DECIMAL(10,2)  
);  
  
CREATE TABLE sales (  
    sales_id INT AUTO_INCREMENT PRIMARY KEY,  
    sales_datetime DATETIME,  
    sales_pos VARCHAR(50),  
    sales_totamount DECIMAL(10,2),  
    sales_discounts DECIMAL(10,2),  
    sales_finalltotal DECIMAL(10,2),  
    sales_status VARCHAR(20),  
    emp_id INT,  
    prod_id INT,  
    cust_id INT,  
    FOREIGN KEY (emp_id) REFERENCES employee(emp_id),  
    FOREIGN KEY (prod_id) REFERENCES product(prod_id),  
    FOREIGN KEY (cust_id) REFERENCES customer(cust_id)  
);
```

Table employee has a connection with table sales. We can see that table sales have emp_id as foreign key which is a primary key in table employee.

10.0 Execution of Data Manipulation Language (DML) Statement – SQL Sets: Questions, SQL Commands, Output, and Number of Records

10.1 Display all records from all tables

Question

Displaying table customer

Screenshot of SQL Command

```
SELECT * FROM customer;
```

Screenshot of Output

cust_id	cust_name	cust_gender	cust_phoneno	cust_email
101	Ahmad bin Hassan	M	0123456789	ahmad.hassan@gmail.com
102	Siti Nurhaliza	F	0198765432	siti.nur@yahoo.com
103	Muhammad Hafiz	M	0167894561	hafiz.m@gmail.com
104	Nur Aisyah	F	0134567890	aisyah.nur@outlook.com
105	Azman bin Osman	M	0145678901	azman.osman@hotmail.com
106	Nurul Ain	F	0156789012	nurul.ain@gmail.com
107	Khairul Anwar	M	0187890123	khairul.a@yahoo.com
108	Fatimah Zahra	F	0129012345	fatimah.z@outlook.com
109	Mohd Faiz	M	0178901234	faiz.mohd@gmail.com
110	Zainab binti Ali	F	0169012345	zainab.ali@hotmail.com
111	Ismail bin Ahmad	M	0190123456	ismail.ahmad@yahoo.com
112	Aishah Maisarah	F	0121234567	aishah.m@gmail.com
113	Haris Fadilah	M	0132345678	haris.f@outlook.com
114	Sarah binti Abdullah	F	0143456789	sarah.abdullah@gmail.com
115	Razak bin Hamid	M	0154567890	razak.hamid@yahoo.com
116	Amirul Hakim	M	0172345678	amirul.hakim@uitm.edu.my
117	Nadira Syafinaz	F	0128765432	nadira.syafinaz@uitm.edu....
118	Fikri Azlan	M	0193456789	fikri.azlan@gmail.com
119	Hani Binti Rosli	F	0135678901	hani.rosli@uitm.edu.my
120	Aizat Bin Zainal	M	0146789012	aizat.zainal@hotmail.com

Screenshot of the Number of Records

```
09:35:57 SELECT * FROM customer LIMIT 0, 1000
```

20 row(s) returned

Question

Displaying table delivery

Screenshot of SQL Command

```
SELECT * FROM delivery;
```

Screenshot of Output

delivery_id	delivery_datetime	delivery_total	delivery_quantity	supp_id	stock_id
501	2025-11-01 09:00:00	1500.00	100	101	301
502	2025-11-03 10:30:00	850.50	50	102	302
503	2025-11-05 08:45:00	2100.00	300	103	303
504	2025-11-07 11:15:00	450.00	90	104	304
505	2025-11-20 14:00:00	350.00	35	901	305
506	2025-11-22 09:30:00	180.00	18	902	306
507	2025-11-25 13:00:00	500.00	50	907	307
508	2025-12-01 08:00:00	3200.00	200	101	308
509	2025-12-03 15:45:00	95.00	10	903	309
510	2025-12-12 10:00:00	400.00	80	103	310
511	2025-11-02 11:00:00	1500.00	120	111	311
512	2025-11-04 09:15:00	850.20	60	112	312
513	2025-11-06 14:30:00	2100.00	150	113	313
514	2025-11-08 08:30:00	450.00	45	114	314
515	2025-12-02 16:00:00	3200.00	100	105	315
516	2025-12-13 11:20:00	400.00	40	115	316
517	2025-11-26 09:00:00	500.00	25	112	317
518	2025-11-21 10:45:00	350.00	10	901	318
519	2025-11-22 13:15:00	350.00	200	904	319
520	2025-11-30 08:30:00	250.00	100	109	320
521	2025-12-04 09:00:00	3200.00	400	121	321
522	2025-12-14 14:30:00	400.00	20	122	322
523	2025-12-05 10:15:00	94.95	15	123	323
524	2025-12-20 08:00:00	800.00	80	201	324
525	2025-12-01 11:00:00	250.00	50	202	325

Screenshot of the Number of Records

```
09:58:03 SELECT * FROM delivery LIMIT 0, 1000
```

25 row(s) returned

Question

Displaying table employee

Screenshot of SQL Command

```
SELECT * FROM employee;
```

Screenshot of Output

emp_id	emp_name	emp_role	emp_phoneno	emp_status	emp_startdate	emp_workhours	emp_certifications	emp_salary
1	Muhammad Zufadli	Manager	0128457391	Active	2022-05-18	9.00	Bachelor's Degree in Business Administration	6500.00
2	Nurul Aini	Assistant Manager 1	0137682945	Active	2022-05-20	8.00	Diploma in Retail Management	3200.00
3	Hassan	Assistant Manager 2	0145938267	Active	2022-06-02	8.00	Certificate in Customer Service & POS Systems	3200.00
4	Hussin	Supervisor	0116748392	Active	2023-03-21	8.00	Certificate in Retail Supervision	2600.00
5	Farah Adilah	Accountant	0163829457	Active	2024-12-09	8.00	Diploma in Accounting	2400.00
6	Nurul Syafie	Part-Time Staff	0174928365	Active	2025-08-30	7.00	Malaysian Certificate of Education	1200.00

Screenshot of the Number of Records

```
10:00:01 SELECT * FROM employee LIMIT 0, 1000 6 row(s) returned
```

Question

Displaying table expense

Screenshot of SQL Command

```
SELECT * FROM expense;
```

Screenshot of Output

exp_id	exp_date	exp_amount	exp_type	exp_status	exp_category	exp_desc	emp_id
1	2025-11-01	1500.00	Purchase	Paid	Inventory	Restock of Ayam Brand and Maggi products	1
2	2025-11-03	850.50	Purchase	Paid	Inventory	Weekly dairy supply from Dutch Lady	2
3	2025-11-05	2100.00	Purchase	Pending	Inventory	Bulk purchase of beverages from Coca Cola	1
4	2025-11-07	450.00	Purchase	Paid	Supplies	Restock of stationery and paper products	3
5	2025-11-10	1200.00	Fixed	Paid	Utilities	TNB Electricity bill for November	4
6	2025-11-10	150.20	Fixed	Paid	Utilities	Syabas Water bill for November	4
7	2025-11-12	300.00	Fixed	Paid	Utilities	Unifi Business Broadband subscription	4
8	2025-11-15	75.00	Claim	Paid	Transport	Petrol claim for supplier meeting	5
9	2025-11-16	120.00	Claim	Approved	Meal	Client lunch meeting at Old Town White Coffee	5
10	2025-11-18	45.00	Claim	Paid	Parking	Site visit parking fees	6
11	2025-11-20	350.00	Variable	Paid	Maintenance	Repairs for store front sliding door	2
12	2025-11-22	180.00	Variable	Paid	Cleaning	Professional pest control service	2
13	2025-11-25	500.00	Variable	Pending	Maintenance	Air-conditioning servicing (3 units)	1
14	2025-11-28	1000.00	Variable	Paid	Marketing	Printing of promotional flyers and banners	3
15	2025-11-30	250.00	Variable	Paid	Staff Welfare	Monthly pantry supplies and staff snacks	6
16	2025-12-01	3200.00	Purchase	Paid	Inventory	Holiday season bulk stock - Snacks & Sweets	1
17	2025-12-02	150.00	Claim	Pending	Postage	Shipping costs for corporate gifts	3
18	2025-12-03	95.00	Variable	Paid	Supplies	Replacement of cleaning tools (mops/buckets)	2
19	2025-12-05	1350.00	Fixed	Pending	Rent	Monthly warehouse storage fee	4
20	2025-12-07	60.00	Claim	Paid	Communication	Monthly mobile phone allowance - Sales Team	5
21	2025-12-10	220.00	Variable	Paid	Safety	Refilling of fire extinguishers	2
22	2025-12-12	400.00	Purchase	Paid	Inventory	Emergency restock of Gardenia bread	1
23	2025-12-15	125.50	Claim	Approved	Transport	Grab car fares for documents delivery	6
24	2025-12-18	55.00	Variable	Paid	Bank Fees	Annual corporate account maintenance fee	4
25	2025-12-20	800.00	Variable	Pending	Marketing	Social media advertisement boost	3

Screenshot of the Number of Records

```
10:01:03 SELECT * FROM expense LIMIT 0, 1000
```

25 row(s) returned

Question

Displaying table payment

Screenshot of SQL Command

```
SELECT * FROM payment;
```

Screenshot of Output

pay_id	pay_method	pay_amount	pay_date	pay_status	supp_id	emp_id
401	Bank Transfer	1500.00	2025-11-02	Completed	101	1
402	Cheque	850.50	2025-11-04	Completed	102	2
403	Bank Transfer	2100.00	2025-11-06	Pending	103	1
404	Credit Card	450.00	2025-11-08	Completed	104	3
405	Bank Transfer	1200.00	2025-11-10	Completed	107	4
406	Cheque	500.00	2025-11-12	Completed	201	2
407	Bank Transfer	350.00	2025-11-14	Completed	301	1
408	Bank Transfer	180.00	2025-11-15	Completed	302	2
409	Credit Card	95.00	2025-11-16	Completed	305	3
410	Bank Transfer	600.00	2025-11-18	Pending	306	4
411	Bank Transfer	1350.00	2025-11-20	Completed	901	4
412	Cheque	400.00	2025-11-22	Completed	902	5
413	Credit Card	220.00	2025-11-24	Completed	904	2
414	Bank Transfer	55.00	2025-11-26	Completed	911	4
415	Cash	150.00	2025-11-28	Completed	913	3
416	Bank Transfer	3200.00	2025-12-01	Completed	101	1
417	Cheque	800.00	2025-12-03	Pending	109	3
418	Bank Transfer	125.50	2025-12-05	Completed	120	6
419	Credit Card	440.00	2025-12-07	Completed	121	5
420	Bank Transfer	900.00	2025-12-10	Completed	205	4
421	Cheque	1100.00	2025-12-12	Completed	401	1
422	Bank Transfer	75.00	2025-12-14	Completed	601	5
423	Cash	30.00	2025-12-15	Completed	701	6
424	Bank Transfer	500.00	2025-12-18	Pending	801	2
425	Credit Card	250.00	2025-12-20	Completed	941	3

Screenshot of the Number of Records

```
10:02:54 SELECT * FROM payment LIMIT 0, 1000
```

25 row(s) returned

Question

Displaying table product

Screenshot of SQL Command

```
SELECT * FROM product;
```

Screenshot of Output

prod_id	prod_name	prod_category	prod_brand	prod_costprice	prod_sellprice
1	Sardine in Tomato 425g	Canned Food	Ayam Brand	3.50	5.00
2	Tuna Chunk 185g	Canned Food	Ayam Brand	4.00	5.50
3	Sweetened Condensed Milk 390g	Canned Food	Marigold	2.80	4.00
4	Tomato Sauce 340g	Canned Food	Maggi	3.00	4.20
5	Baked Beans 230g	Canned Food	Ayam Brand	3.50	5.00
6	Sardine in Tomato 230g	Canned Food	King Cup	2.50	3.50
7	Fresh Milk 1L	Canned Food	Farm Fresh	5.00	7.00
8	Peach Slices 825g	Canned Food	Delmonte	6.00	8.50
9	Sweet Corn 340g	Canned Food	Ayam Brand	3.00	4.50
10	Oyster Sauce 510g	Canned Food	Lee Kum Kee	5.00	7.00
11	Fresh Milk 1L	Dairy	Dutch Lady	5.50	7.50
12	Milk Powder 900g	Dairy	Anlene	22.00	28.00
13	Yogurt 130g	Dairy	Nestle	1.80	2.50
14	Butter 250g	Dairy	Buttercup	4.50	6.00
15	Cheddar Cheese 180g	Dairy	Kraft	8.00	11.00
16	Fresh Milk 1L	Dairy	Farm Fresh	6.50	9.00
17	Cream Cheese 250g	Dairy	Philadelphia	10.00	14.00
18	Cultured Milk 5 pack	Dairy	Vitagen	4.00	5.50
19	Teh Tarik 15 sachets	Beverages	Karak Wangi	3.00	4.50
20	Black Coffee 20 sachets	Beverages	Old Town	6.00	8.50
21	Chocolate Malt Drink 1kg	Beverages	Milo	18.00	24.00
22	3in1 Coffee 20 sachets	Beverages	Nescafe	7.50	10.00
23	Mineral Water 1.5L	Beverages	Spritzer	1.00	1.50
24	Soy Bean Drink 1L	Beverages	Drinho	2.00	3.00
25	Blackcurrant Cordial 2L	Beverages	Ribena	8.00	11.00
26	Carbonated Drink 325ml	Beverages	Coca Cola	1.50	2.00

Screenshot of the Number of Records

10:04:15 SELECT * FROM product LIMIT 0, 1000

139 row(s) returned

Question

Displaying table sales

Screenshot of SQL Command

```
SELECT * FROM sales;
```

Screenshot of Output

sales_id	sales_datetime	sales_pos	sales_totamount	sales_discounts	sales_finalltotal	sales_status	emp_id	prod_id	cust_id
901	2025-11-01 10:15:00	Register 1	45.00	0.00	45.00	Completed	1	23	102
902	2025-11-01 11:30:00	Register 2	120.50	5.50	115.00	Completed	2	121	110
903	2025-11-01 14:45:00	Register 1	15.00	0.00	15.00	Completed	1	105	119
904	2025-11-02 09:20:00	Mobile Tablet	85.00	10.00	75.00	Completed	3	11	120
905	2025-11-02 12:05:00	Register 2	32.00	0.00	32.00	Completed	2	26	101
906	2025-11-03 16:40:00	Register 1	55.20	2.20	53.00	Completed	1	89	101
907	2025-11-03 18:10:00	Register 1	10.00	0.00	10.00	Cancelled	4	45	115
908	2025-11-04 11:00:00	Register 2	200.00	20.00	180.00	Completed	2	101	114
909	2025-11-04 13:25:00	Mobile Tablet	65.00	5.00	60.00	Completed	3	21	106
910	2025-11-05 15:50:00	Register 1	42.00	0.00	42.00	Completed	5	99	109
911	2025-11-06 10:05:00	Register 2	25.00	0.00	25.00	Completed	2	72	116
912	2025-11-06 14:30:00	Register 1	98.00	8.00	90.00	Completed	1	16	111
913	2025-11-07 09:45:00	Mobile Tablet	30.00	3.00	27.00	Completed	6	9	101
914	2025-11-07 17:15:00	Register 2	50.00	0.00	50.00	Completed	4	115	104
915	2025-11-08 12:20:00	Register 1	15.50	0.00	15.50	Completed	5	82	115
916	2025-11-08 19:00:00	Register 1	110.00	10.00	100.00	Completed	1	111	116
917	2025-11-09 11:30:00	Register 2	45.00	0.00	45.00	Completed	2	101	113
918	2025-11-09 14:10:00	Mobile Tablet	22.00	0.00	22.00	Completed	3	52	101
919	2025-11-10 10:40:00	Register 1	75.00	5.00	70.00	Completed	5	79	106
920	2025-11-11 16:20:00	Register 2	12.00	0.00	12.00	Completed	2	113	103
921	2025-11-12 09:55:00	Register 1	140.00	15.00	125.00	Completed	1	71	120
922	2025-11-12 13:15:00	Mobile Tablet	38.00	0.00	38.00	Completed	3	90	118
923	2025-11-13 15:45:00	Register 2	60.00	5.00	55.00	Completed	4	74	112
924	2025-11-14 11:20:00	Register 1	28.00	0.00	28.00	Completed	5	66	114
925	2025-11-15 18:30:00	Register 1	250.00	25.00	225.00	Completed	1	11	106

Screenshot of the Number of Records

```
10:06:58 SELECT * FROM sales LIMIT 0, 1000 25 row(s) returned
```


Question

Displaying table stock

Screenshot of SQL Command

```
SELECT * FROM stock;
```

Screenshot of Output

stock_id	stock_quantity	stock_unitcost	stock_totalcost	stock_status	prod_id	exp_id
301	100	15.00	1500.00	In Stock	1	1
302	50	17.01	850.50	In Stock	23	2
303	300	7.00	2100.00	In Stock	113	3
304	90	5.00	450.00	Low Stock	34	4
305	35	10.00	350.00	In Stock	83	11
306	18	10.00	180.00	Low Stock	91	12
307	50	10.00	500.00	In Stock	111	13
308	200	16.00	3200.00	In Stock	2	16
309	10	9.50	95.00	In Stock	45	18
310	80	5.00	400.00	In Stock	10	22
311	120	12.50	1500.00	In Stock	11	1
312	60	14.17	850.20	In Stock	99	2
313	150	14.00	2100.00	Reserved	73	3
314	45	10.00	450.00	Low Stock	41	4
315	100	32.00	3200.00	In Stock	22	16
316	40	10.00	400.00	In Stock	102	22
317	25	20.00	500.00	In Stock	109	13
318	10	35.00	350.00	Low Stock	97	11
319	200	1.75	350.00	In Stock	121	11
320	100	2.50	250.00	In Stock	100	15
321	400	8.00	3200.00	In Stock	1	16
322	20	20.00	400.00	In Stock	12	22
323	15	6.33	94.95	Low Stock	113	18
324	80	10.00	800.00	In Stock	104	25
325	50	5.00	250.00	In Stock	67	15

Screenshot of the Number of Records

```
10:08:01 SELECT * FROM stock LIMIT 0, 1000
```

25 row(s) returned

Question

Displaying table supplier

Screenshot of SQL Command

```
SELECT * FROM supplier;
```

Screenshot of Output

supp_id	supp_name	supp_phoneno	supp_email	supp_unithum	supp_street	supp_neighborhood	supp_town	supp_state
101	Ayam Brand Marketing Sdn Bhd	0378945601	sales@ayambrand.com.my	Lot 15	Jalan Industri 3	Taman Perindustrian Bukit Raja	Shah Alam	Selangor
102	Marigold HL Malaysia Sdn Bhd	0167823402	inquiry@marigold.com.my	No 8	Jalan Klang	Taman Perindustrian Bolton	Petaling Jaya	Selangor
103	Nestle Manufacturing Malaysia	0355678903	contact@nestle.com.my	No 22-1	Jalan Subang 5	Subang Industrial Park	Subang Jaya	Selangor
104	Maggi Food Products Malaysia	0198234504	sales@maggi.com.my	Lot 22	Jalan Perusahaan 4	Kawasan Perindustrian Meru	Klang	Selangor
105	Delmonte Foods Malaysia Sdn Bhd	0145672305	info@delmonte.com.my	No 45	Jalan Perusahaan Satu	Taman Industri Puchong	Puchong	Selangor
106	Lee Kum Kee Malaysia Sdn Bhd	0387651206	sales@lkk.com.my	Unit 12	Jalan Reko	Taman Industri	Kajang	Selangor
107	Dutch Lady Milk Industries	0356789407	inquiry@dutchlady.com.my	No 1	Jalan Pendidikan	Section 15	Petaling Jaya	Selangor
108	Anlene Marketing Sdn Bhd	0321234508	contact@anlene.com.my	Level 8	Jalan Imbi	Bukit Bintang	Kuala Lumpur	Wilayah Persekutuan
109	Kraft Foods Malaysia	0178923409	sales@kraft.com.my	No 3	Jalan Utama 2/1	Taman Perindustrian Puchong Utama	Shah Alam	Selangor
110	Philadelphia Trading Malaysia	0134561210	info@philadelphia.com.my	Unit 5-2	Jalan SS15/4D	Kompleks Perniagaan	Subang Jaya	Selangor
111	Farm Fresh Marketing Sdn Bhd	0367892311	sales@farmfresh.com.my	Lot 2A	Jalan Sungai Buloh	Kampung Baru Sungai Buloh	Sungai Buloh	Selangor
112	Vitagen Malaysia Sdn Bhd	0189456712	inquiry@vitagen.com.my	No 18	Jalan TPP 1/3	Taman Perindustrian Puchong	Puchong	Selangor
113	Karak Wangi Food Industries	0156783413	contact@karakwangi.com....	Lot 35	Jalan Perusahaan 6	Kawasan Perindustrian Rawang	Rawang	Selangor
114	Old Town White Coffee Sdn Bhd	0342156714	sales@oldtown.com.my	No 288	Jalan Ipoh	Taman Wahyu	Kuala Lumpur	Wilayah Persekutuan
115	Milo Malaysia Distribution	0398761215	info@milo.com.my	Lot 6	Jalan Perindustrian Bu...	Shah Alam Industrial Park	Shah Alam	Selangor
116	Nescafe Malaysia Sdn Bhd	0167234516	contact@nescafe.com.my	No 12-3	Jalan PJU 1/43	Dataran Prima	Petaling Jaya	Selangor
117	Spritzer Malaysia Sdn Bhd	0545678917	sales@spritzer.com.my	Lot 1001	Jalan Kuala Bikam	Kampung Tersusun	Taiping	Perak
118	Drinho Food Industries	0198723418	inquiry@drinho.com.my	No 25	Jalan Kapar 27/89	Taman Klang Jaya	Klang	Selangor
119	Ribena Malaysia Distributors	0335674519	contact@ribena.com.my	No 18-2	Jalan Serai Wangi	Taman Perindustrian UEP	Shah Alam	Selangor
120	Coca Cola Malaysia Sdn Bhd	0145623920	sales@cocacola.com.my	Lot 9	Jalan Astaka U8/84	Seksyen U8 Bukit Jelutong	Petaling Jaya	Selangor
121	F&N Marketing Malaysia	0378912321	info@fn.com.my	No 33	Jalan TPP 5/15	Taman Perindustrian Puchong	Puchong	Selangor
122	Hales Marketing Sdn Bhd	0189234522	contact@hales.com.my	Lot 12	Jalan Semenyih Sentr...	Taman Industri Semenyih	Kajang	Selangor
123	Horlicks Distribution Malaysia	0356781223	sales@horlicks.com.my	Unit 2-8	Jalan SS9/19	Seksyen 9	Subang Jaya	Selangor
201	Maggi Instant Noodles Malaysia	0167893424	noodles@maggi.com.my	Lot 28	Jalan Perusahaan Empat	Kawasan Perindustrian Meru	Shah Alam	Selangor
202	Mee Sedap Food Industries	0333456725	sales@meesedap.com.my	No 45	Jalan Kapar Batu 7	Taman Klang Utama	Klang	Selangor
203	Cintan Noodles Malaysia	0198567826	info@cintan.com.my	Lot 88	Jalan Rawang Selayang	Taman Perindustrian Selayang	Rawang	Selangor

Screenshot of the Number of Records

```
10:08:54 SELECT * FROM supplier LIMIT 0, 1000
```

104 row(s) returned

10.2 List significant information from a single table

Question

Display customer's id, customer's name, customer's phone number from table customer.

Screenshot of SQL Command

```
USE retail_shop;  
SELECT cust_id, cust_name, cust_phoneno  
FROM customer;
```

Screenshot of Output

cust_id	cust_name	cust_phoneno
101	Ahmad bin Hassan	0123456789
102	Siti Nurhaliza	0198765432
103	Muhammad Hafiz	0167894561
104	Nur Aisyah	0134567890
105	Azman bin Osman	0145678901
106	Nurul Ain	0156789012

Screenshot of the Number of Records

17:04:44 SELECT cust_id, cust_name, cust_phoneno FROM customer LIMIT 0, 50

20 row(s) returned

Question

Display product's id, product's name and product's sell price from table product.

Screenshot of SQL Command

```
USE retail_shop;  
SELECT prod_id, prod_name, prod_sellprice  
FROM product;
```

Screenshot of Output

prod_id	prod_name	prod_sellprice
1	Sardine in Tomato 425g	5.00
2	Tuna Chunk 185g	5.50
3	Sweetened Condensed Milk 390g	4.00
4	Tomato Sauce 340g	4.20
5	Baked Beans 230g	5.00
6	Sardine in Tomato 230g	3.50

Screenshot of the Number of Records

17:07:51 SELECT prod_id, prod_name, prod_sellprice FROM product LIMIT 0, 50

50 row(s) returned

10.3 List significant information that uses a comparison operator

Question

Displaying employee's name, employee's start date and employee's role from table employee for employee that start before 2024 in ascending order.

Screenshot of SQL Command

```
USE retail_shop;  
SELECT * from employee;  
SELECT emp_name, emp_startdate, emp_role  
FROM employee  
WHERE emp_startdate < '2024-01-01'  
ORDER BY emp_startdate ASC;
```

Screenshot of Output

emp_name	emp_startdate	emp_role
Muhammad Zulfadli	2022-05-18	Manager
Nurul Aini	2022-05-20	Assistant Manager 1
Hassan	2022-06-02	Assistant Manager 2
Hussin	2023-03-21	Supervisor

Screenshot of the Number of Records

17:22:26 SELECT emp_name, emp_startdate, emp_role FROM employee WHERE emp_startdate < '2024-01-01' ORDER BY em... 4 row(s) returned

Question

Displaying product's id, product's name and product's sell price from table product where the sell price is more than 5.

Screenshot of SQL Command

```
USE retail_shop;  
SELECT prod_id, prod_name, prod_sellprice  
FROM product  
WHERE prod_sellprice > 5;
```

Screenshot of Output

prod_id	prod_name	prod_sellprice
2	Tuna Chunk 185g	5.50
7	Fresh Milk 1L	7.00
8	Peach Slices 825g	8.50
10	Oyster Sauce 510g	7.00
11	Fresh Milk 1L	7.50
12	Milk Powder 900g	28.00
14	Butter 250g	6.00
15	Cheddar Cheese 180g	11.00
16	Fresh Milk 1L	9.00
17	Cream Cheese 250g	14.00
18	Cultured Milk 5 pack	5.50
20	Black Coffee 20 sachets	8.50
21	Chocolate Malt Drink ...	24.00
22	3in1 Coffee 20 sachets	10.00
25	Blackcurrant Cordial 2L	11.00
29	Malted Drink 500g	20.00
34	Fried Noodles 5 pack	5.50
37	Spicy Noodles Cup	6.00
52	Sweet Soy Sauce 645ml	5.50
54	Mayonnaise 470ml	7.00
55	Curry Powder 250g	6.00
61	Fried Shallots 100g	6.00
62	Grade A Eggs 10 pcs	6.50
63	Grade B Eggs 10 pcs	5.50
64	Omega Eggs 10 pcs	8.00
69	Donuts 6 pack	7.00

Screenshot of the Number of Records

```
17:09:52 SELECT prod_id, prod_name, prod_sellprice FROM product WHERE prod_sellprice > 5 LIMIT 0, 50 50 row(s) returned
```

10.4 List significant information that uses “IN” and “LIKE” clauses

Question

Displaying product's name, product's sell price and product's category from the table product where the product's category is “Beverages”. The list is in descending order of the sell price.

Screenshot of SQL Command

```
USE retail_shop;  
SELECT prod_name, prod_sellprice, prod_category  
FROM product  
WHERE prod_category LIKE '%Beverages%'  
ORDER BY prod_sellprice DESC;
```

Screenshot of Output

prod_name	prod_sellprice	prod_category
Chocolate Malt Drink 1kg	24.00	Beverages
Malted Drink 500g	20.00	Beverages
Blackcurrant Cordial 2L	11.00	Beverages
3in1 Coffee 20 sachets	10.00	Beverages
Black Coffee 20 sachets	8.50	Beverages
Teh Tarik 15 sachets	4.50	Beverages

Screenshot of the Number of Records

```
17:26:56 SELECT prod_name, prod_sellprice, prod_category FROM product WHERE prod_category LIKE '%Beverages%' ORDE... 11 row(s) returned
```

Question

Displaying product's id, product's name, stock quantity and product's sell price from table product and stock where the product's brand is "Ayam Brand" and "Maggi". The list is in descending order of the sell price.

Screenshot of SQL Command

```
SELECT p.prod_id, p.prod_name, p.prod_brand, s.stock_quantity, p.prod_sellprice
FROM product p
LEFT JOIN stock s ON p.prod_id = s.prod_id
WHERE p.prod_brand IN ('Ayam Brand', 'Maggi')
ORDER BY p.prod_sellprice DESC;
```

Screenshot of Output

prod_id	prod_name	prod_brand	stock_quantity	prod_sellprice
2	Tuna Chunk 185g	Ayam Brand	200	5.50
1	Sardine in Tomato 425g	Ayam Brand	100	5.00
1	Sardine in Tomato 425g	Ayam Brand	400	5.00
5	Baked Beans 230g	Ayam Brand	150	5.00
30	Curry Flavour Noodles 5 pack	Maggi	120	5.00
31	Asam Laksa Noodles 5 pack	Maggi	120	5.00
35	Tom Yam Noodles 5 pack	Maggi	120	5.00
53	BBQ Sauce 340g	Maggi	85	5.00
9	Sweet Corn 340g	Ayam Brand	200	4.50
4	Tomato Sauce 340g	Maggi	180	4.20

Screenshot of the Number of Records

11:16:44 SELECT p.prod_id, p.prod_name, p.prod_brand, s.stock_quantity, p.prod_sellprice FROM product p LEFT JOIN stock s ON p.prod_id = s.prod_id WHE... 10 row(s) returned

10.5 List significant information that uses the column functions clause

Question

Determine the total amount of expenses approved by each employee.
Display the employee's name, role, and phone number.

Screenshot of SQL Command

```
SELECT emp_name, emp_role, emp_phoneno, SUM(exp_amount) as TotalExpense
FROM employee
JOIN expense ON employee.emp_id = expense.emp_id
GROUP BY emp_name, emp_role, emp_phoneno;
```

Screenshot of Output

emp_name	emp_role	emp_phoneno	TotalExpense
Muhammad Zulfadli	Manager	0128457391	7700.00
Nurul Aini	Assistant Manager 1	0137682945	1695.50
Hassan	Assistant Manager 2	0145938267	2400.00
Hussin	Supervisor	0116748392	3055.20
Farah Adilah	Accountant	0163829457	255.00
Nurul Syafie	Part-Time Staff	0174928365	420.50

Screenshot of the Number of Records

```
21:48:56 SELECT emp_name, emp_role, emp_phoneno, SUM(exp_amount) as TotalExpense FROM employee JOIN expense ON employee.emp_id = expense.e... 6 row(s) returned
```

Question

Find the average stock quantity for each product. List the product names along with their average stock.

Screenshot of SQL Command

```
SELECT prod_name, avg(stock_quantity)
FROM stock
JOIN product ON stock.prod_id = product.prod_id
GROUP BY prod_name;
```

Screenshot of Output

prod_name	avg(stock_quantity)
Sardine in Tomato 425g	250.0000
Mineral Water 1.5L	50.0000
Cooling Powder 50g	157.5000
Fried Noodles 5 pack	90.0000
Detergent Powder 2.5kg	35.0000
Toothpaste 225g	18.0000
Correction Tape	50.0000
Tuna Chunk 185g	200.0000
Frozen Paratha 5 pack	10.0000
Oyster Sauce 510g	80.0000
Fresh Milk 1L	120.0000
Toilet Roll 10 pack	60.0000
Chicken Franks 340g	150.0000
Onion Ring Snack 60g	45.0000
3in1 Coffee 20 sachets	100.0000
Baby Powder 400g	40.0000
Eraser	25.0000
Hair Conditioner 340ml	10.0000
Chocolate Stick	200.0000
Kitchen Towel 2 rolls	100.0000
Milk Powder 900g	20.0000
Baby Lotion 200ml	80.0000
Wholemeal Bread 400g	50.0000
Baked Beans 230g	150.0000
Curry Flavour Noodles ...	120.0000
Asam Laksa Noodles 5 p...	120.0000

Screenshot of the Number of Records

```
21:51:20 SELECT prod_name, avg(stock_quantity) FROM stock JOIN product ON stock.prod_id = product.prod_id GROUP BY prod_name LIMIT 0, 1000
```

30 row(s) returned

10.6 List significant information that uses the column functions clause with the “HAVING” clause

Question

Identify the payment method whose average payment amount is higher than the total amount.

Screenshot of SQL Command

```
SELECT pay_method, AVG(pay_amount) as PayAmount
FROM payment
GROUP BY pay_method
HAVING PayAmount > (SELECT AVG(pay_amount) FROM payment);
```

Screenshot of Output

pay_method	PayAmount
Bank Transfer	933.500000
Cheque	730.100000

Screenshot of the Number of Records

```
21:58:56 SELECT pay_method, AVG(pay_amount) as PayAmount FROM payment GROUP BY pay_method HAVING PayAmount > (SELECT AVG(pay_amount) ... 2 row(s) returned
```

Question

List the employees who have recorded total sales exceeding RM500.

Screenshot of SQL Command

```
SELECT emp_id, SUM(sales_totamount) as TotalSales
FROM sales
GROUP BY emp_id
HAVING TotalSales > 500;
```

Screenshot of Output

emp_id	TotalSales
1	713.20

Screenshot of the Number of Records

22:02:49 SELECT emp_id, SUM(sales_totamount) as TotalSales FROM sales GROUP BY emp_id HAVING TotalSales > 500 LIMIT 0, 1000 1 row(s) returned

10.7 List significant information that uses the subquery clause

Question

Identify the suppliers that have not been contacted by the owner during the specified month.

Screenshot of SQL Command

```
SELECT supp_name
FROM supplier
WHERE supp_id NOT IN (SELECT supp_id FROM payment);
```

Screenshot of Output

supp_name
Delmonte Foods Malaysia Sdn Bhd
Lee Kum Kee Malaysia Sdn Bhd
Anlene Marketing Sdn Bhd
Philadelphia Trading Malaysia
Farm Fresh Marketing Sdn Bhd
Vitagen Malaysia Sdn Bhd
Karak Wangi Food Industries
Old Town White Coffee Sdn Bhd
Milo Malaysia Distribution
Nescafe Malaysia Sdn Bhd
Spritzer Malaysia Sdn Bhd
Drinho Food Industries
Ribena Malaysia Distributors
Hales Marketing Sdn Bhd
Horlicks Distribution Malaysia
Mee Sedap Food Industries
Cintan Noodles Malaysia
Indomie Malaysia Sdn Bhd
Twisties Snack Foods Malaysia
Super Ring Foods Malaysia
Kawan Food Manufacturing
Munchy Food Industries
Life Brand Marketing Malaysia
Agricare Products Malaysia
Cap Payung Industries Sdn Bhd
MSM Malaysia Holdings

Screenshot of the Number of Records

22:13:14 SELECT supp_name FROM supplier WHERE supp_id NOT IN (SELECT supp_id FROM payment) LIMIT 0, 1000

80 row(s) returned

Question

Determine the highest sales record along with the employee who made the sale.

Screenshot of SQL Command

```
SELECT emp_id, sales_totamount
FROM sales
WHERE sales_totamount = (SELECT MAX(sales_totamount) FROM sales);
```

Screenshot of Output

emp_id	sales_totamount
1	250.00

Screenshot of the Number of Records

```
22:14:52 SELECT emp_id, sales_totamount FROM sales WHERE sales_totamount = (SELECT MAX(sales_totamount) FROM sales) LIMIT 0, 1000 1 row(s) returned
```

10.8 List significant information that uses the join clause with other tables

Question

Display the names and phone numbers of customers along with their total purchase amounts.

Screenshot of SQL Command

```
SELECT cust_name, cust_phoneno, sales_finaltotal, sales_discounts
FROM customer
JOIN sales ON customer.cust_id = sales.cust_id;
```

Screenshot of Output

cust_name	cust_phoneno	sales_finaltotal	sales_discounts
Siti Nurhaliza	0198765432	45.00	0.00
Zainab binti Ali	0169012345	115.00	5.50
Hani Binti Rosli	0135678901	15.00	0.00
Aizat Bin Zainal	0146789012	75.00	10.00
Ahmad bin Hassan	0123456789	32.00	0.00
Ahmad bin Hassan	0123456789	53.00	2.20
Razak bin Hamid	0154567890	10.00	0.00
Sarah binti Abdullah	0143456789	180.00	20.00
Nurul Ain	0156789012	60.00	5.00
Mohd Faiz	0178901234	42.00	0.00
Amirul Hakim	0172345678	25.00	0.00
Ismail bin Ahmad	0190123456	90.00	8.00
Ahmad bin Hassan	0123456789	27.00	3.00
Nur Aisyah	0134567890	50.00	0.00
Razak bin Hamid	0154567890	15.50	0.00
Amirul Hakim	0172345678	100.00	10.00
Haris Fadhillah	0132345678	45.00	0.00
Ahmad bin Hassan	0123456789	22.00	0.00
Nurul Ain	0156789012	70.00	5.00
Muhammad Hafiz	0167894561	12.00	0.00
Aizat Bin Zainal	0146789012	125.00	15.00
Fikri Azlan	0193456789	38.00	0.00
Aishah Maisarah	0121234567	55.00	5.00
Sarah binti Abdullah	0143456789	28.00	0.00
Nurul Ain	0156789012	225.00	25.00

Screenshot of the Number of Records

22:18:04 SELECT cust_name, cust_phoneno, sales_finaltotal, sales_discounts FROM customer JOIN sales ON customer.cust_id = sales.cust_id LIMIT 0, 1000 25 row(s) returned

Question

List the suppliers' details along with their delivery dates.

Screenshot of SQL Command

```
SELECT supp_name, supp_phoneno, supp_street, delivery_datetime  
FROM supplier  
JOIN delivery ON supplier.supp_id = delivery.supp_id;
```

Screenshot of Output

supp_name	supp_phoneno	supp_street	delivery_datetime
Ayam Brand Marketing Sdn Bhd	0378945601	Jalan Industri 3	2025-11-01 09:00:00
Marigold HL Malaysia Sdn Bhd	0167823402	Jalan Kilang	2025-11-03 10:30:00
Nestle Manufacturing Malaysia	0355678903	Jalan Subang 5	2025-11-05 08:45:00
Maggi Food Products Malaysia	0198234504	Jalan Perusahaan 4	2025-11-07 11:15:00
Mama Lemon Malaysia Sdn Bhd	0387654557	Jalan TPP 3/6	2025-11-20 14:00:00
Breeze Detergent Malaysia	0334123458	Jalan Kapar Batu 10	2025-11-22 09:30:00
Magiclean Products Malaysia	0167823463	Jalan TPP 5/12	2025-11-25 13:00:00
Ayam Brand Marketing Sdn Bhd	0378945601	Jalan Industri 3	2025-12-01 08:00:00
SC Johnson Malaysia (Mr Muscle)	0356789159	Jalan PJU 1/37	2025-12-03 15:45:00
Nestle Manufacturing Malaysia	0355678903	Jalan Subang 5	2025-12-12 10:00:00
Farm Fresh Marketing Sdn Bhd	0367892311	Jalan Sungai Buloh	2025-11-02 11:00:00
Vitagen Malaysia Sdn Bhd	0189456712	Jalan TPP 1/3	2025-11-04 09:15:00
Karak Wangi Food Industries	0156783413	Jalan Perusahaan 6	2025-11-06 14:30:00
Old Town White Coffee Sdn Bhd	0342156714	Jalan Ipoh	2025-11-08 08:30:00
Delmonte Foods Malaysia Sdn Bhd	0145672305	Jalan Perusahaan ...	2025-12-02 16:00:00
Milo Malaysia Distribution	0398761215	Jalan Perindustrian...	2025-12-13 11:20:00
Vitagen Malaysia Sdn Bhd	0189456712	Jalan TPP 1/3	2025-11-26 09:00:00
Mama Lemon Malaysia Sdn Bhd	0387654557	Jalan TPP 3/6	2025-11-21 10:45:00
Dettol Reckitt Malaysia	0198567860	Jalan Perusahaan 4	2025-11-22 13:15:00
Kraft Foods Malaysia	0178923409	Jalan Utama 2/1	2025-11-30 08:30:00
F&N Marketing Malaysia	0378912321	Jalan TPP 5/15	2025-12-04 09:00:00
Hales Marketing Sdn Bhd	0189234522	Jalan Semenyih Se...	2025-12-14 14:30:00
Horlicks Distribution Malaysia	0356781223	Jalan SS9/19	2025-12-05 10:15:00
Maggi Instant Noodles Malaysia	0167893424	Jalan Perusahaan ...	2025-12-20 08:00:00
Mee Sedap Food Industries	0333456725	Jalan Kapar Batu 7	2025-12-01 11:00:00

Screenshot of the Number of Records

```
22:19:48 SELECT supp_name, supp_phoneno, supp_street, delivery_datetime FROM supplier JOIN delivery ON supplier.supp_id = delivery.supp_id LIMIT 0, 1000 25 row(s) returned
```


11.0 Reference

1. Connolly, T., & Begg, C. (2015). Database systems: A practical approach to design, implementation, and management (6th ed.). Pearson.
2. Coronel, C., & Morris, S. (2018). Database systems: Design, implementation, & management (13th ed.). Cengage Learning.
3. Elmasri, R., & Navathe, S. B. (2016). Fundamentals of database systems (7th ed.). Pearson.
4. Pratt, P. J., & Last, M. Z. (2014). Concepts of database management (8th ed.). Cengage Learning.
5. Programming with Mosh. (2019, July 15). SQL tutorial for beginners: Learn SQL in 1 hour [Video]. YouTube. <https://www.youtube.com/watch?v=HXV3zeQKqGY>
6. freeCodeCamp.org. (2018, December 4). Database design course – Learn how to design and plan a database [Video]. YouTube. <https://www.youtube.com/watch?v=ztHopE5Wnpc>
7. Interview with Encik Zulfadli, owner of Gerai 24, conducted face-to-face on October 20, 2025.

12.0 Appendices



The front of Pasar Mini Mohd Salleh



Products that are sold in Pasar Mini Mohd Salleh



Discounts at Pasar Mini Mohd Salleh



Beverages sold in Pasar Mini Mohd Salleh